

MATB43 · Tutorial 3

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1. Motivation: from $\mathbb{N}$ to $\mathbb{R}$

$$\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$$

Real analysis depends on the properties of $\mathbb{R}$. Constructing the real numbers from scratch gives a rigorous foundation for the definitions that follow.

1.1. Recall: extending the number systems

We start with $\mathbb{N} = \{1, \hat{1}, \hat{1}, \dots\}$ and recursively define addition and multiplication.

- **Integers allow subtraction.** Identify pairs with the same difference:

$$(a, b) \sim (c, d) \Leftrightarrow a + d = b + c.$$

Intuitively, this says $a - b = c - d$.

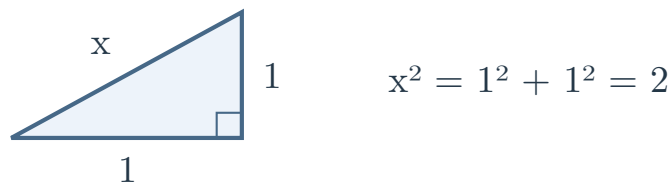
- **Rationals allow division by nonzero numbers.** The equation $2x = 1$ has no integer solution. For integer pairs with nonzero second coordinates, set

$$(a, b) \sim (c, d) \Leftrightarrow ad = bc.$$

Intuitively, this says $\frac{a}{b} = \frac{c}{d}$.

1.2. Why construct the real numbers?

The equation $x^2 = 2$ has no rational solution, although a right triangle with legs of length 1 has a diagonal of precisely this length.



We want to fill the gaps in $\mathbb{Q}$ and obtain a complete ordered number system. For example, consider the rational lower set

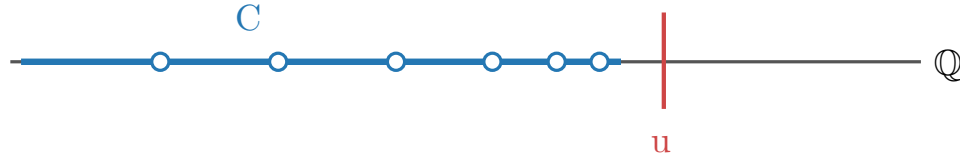
$$S = \{a \in \mathbb{Q} : a < 2\}.$$

It has no greatest element: given $a < 2$, the rational $\frac{a+2}{2}$ lies strictly between a and 2.

2. Dedekind cuts

Definition 2.1 (Dedekind cut). A subset $C \subseteq \mathbb{Q}$ is a **Dedekind cut** if:

1. **Nonempty:** $C \neq \emptyset$.
2. **Downward closed:** if $s \in C$ and $t \in \mathbb{Q}$ with $t < s$, then $t \in C$.
3. **No greatest element:** for every $s \in C$, there is a $t \in C$ with $s < t$.
4. **Bounded above in $\mathbb{Q}$:** there is a $u \in \mathbb{Q}$ such that $s < u$ for all $s \in C$.



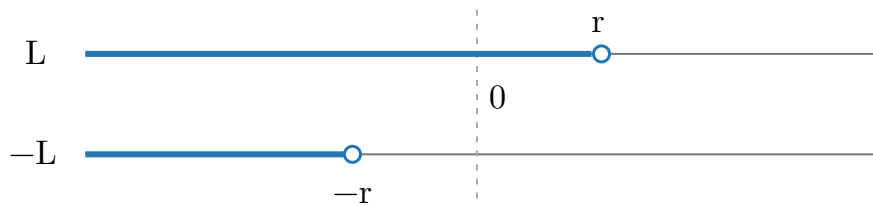
Example. For every $u \in \mathbb{Q}$, the set $C_u := \{s \in \mathbb{Q} : s < u\}$ is a cut. These cuts will represent the rational numbers inside $\mathbb{R}$.

2.1. Negatives: Problem set 4

If L is a cut, define its negative by

$$-L := \{c \in \mathbb{Q} : \exists r \in \mathbb{Q} \setminus L \text{ } c < -r\}.$$

The strict inequality is essential: simply reflecting the complement can produce a greatest element. For instance, $-C_u = C_{-u}$.



2.2. The real numbers and their order

Definition 2.2.

$$\mathbb{R} := \{C : C \text{ is a Dedekind cut of } \mathbb{Q}\}.$$

For cuts C, D , define $C \leq D$ if and only if $C \subseteq D$.

Proposition 2.3. Any two cuts are comparable: $C \subseteq D$ or $D \subseteq C$.

Proof. Suppose $C \not\subseteq D$. Choose $r \in C \setminus D$. If $d \in D$, then $d < r$: equality contradicts $r \notin D$, and $d > r$ would put r in D by downward closure. Since $r \in C$, downward closure of C gives $d \in C$. Thus $D \subseteq C$. ■

Inclusion is reflexive, antisymmetric, transitive, and comparable, so this is a total order.

2.3. Addition and multiplication

For arbitrary cuts, define

$$C + D := \{c + d : c \in C, d \in D\}.$$

For **positive cuts** $C > C_0$ and $D > C_0$, define

$$CD := \{q \in \mathbb{Q} : \exists c \in C \exists d \in D \text{ } c > 0 \text{ and } d > 0 \text{ and } q < cd\}.$$

Define multiplication by zero separately; use negatives for other signs.

Exercise 2.4. Check that these sets are cuts.

3. Constructing $\sqrt{2}$ as a cut

Let

$$S := \{s \in \mathbb{Q} : s < 0 \text{ or } s^2 < 2\}.$$

We will show that S is a positive cut and that $SS = C_2$.

3.1. Checking the cut conditions

Proof.

1. **Nonempty:** $0 \in S$, since $0^2 < 2$.
2. **Bounded above:** 2 is an upper bound outside S . A nonnegative member s has $s^2 < 2$, so it cannot satisfy $s \geq 2$.
3. **Downward closed:** let $s \in S$ and $r < s$. If $r < 0$, then $r \in S$. Otherwise $0 \leq r < s$, so $r^2 < s^2 < 2$ and again $r \in S$.
4. **No greatest element:** if $b \in S$ and $b < 0$, then $0 \in S$ is larger. If $b \geq 0$, let $\delta = 2 - b^2 > 0$ and choose a rational

$$0 < \varepsilon < \min\left(1, \frac{\delta}{2b+1}\right).$$

Then

$$(b + \varepsilon)^2 = b^2 + 2b\varepsilon + \varepsilon^2 < b^2 + (2b + 1)\varepsilon < 2.$$

Thus $b + \varepsilon \in S$ and $b + \varepsilon > b$. ■

Since $C_0 \subset S$ and $0 \in S \setminus C_0$, the cut S is positive.

3.2. Showing $SS \subseteq C_2$

Proof. Let $q \in SS$. If $q \leq 0$, then $q < 2$, so $q \in C_2$. If $q > 0$, the product definition gives positive $a, b \in S$ with $q < ab$. By totality of the rational order, we may assume $a \geq b$. Hence

$$q < ab \leq a^2 < 2,$$

so $q \in C_2$. ■

3.3. Showing $C_2 \subseteq SS$

Proof. Let $c \in C_2$, so $c < 2$. If $c \leq 0$, then $1 \in S$ and $c < 1 \cdot 1$, which puts c in SS .

Now suppose $0 < c < 2$. We construct positive rationals whose squares are arbitrarily close to 2 from below. Define

$$x_1 = 1, \quad x_{n+1} = \frac{2x_n + 2}{x_n + 2}.$$

If $1 \leq x_n$ and $x_n^2 < 2$, then

$$x_{n+1} - x_n = \frac{2 - x_n^2}{x_n + 2} > 0,$$

and direct calculation gives

$$2 - x_{n+1}^2 = \frac{2(2 - x_n^2)}{(x_n + 2)^2}.$$

Therefore $x_{n+1} > x_n \geq 1$ and $x_{n+1}^2 < 2$. Induction proves these properties for every n . Moreover, $(x_n + 2)^2 \geq 9$, so

$$0 < 2 - x_{n+1}^2 \leq \frac{2}{9}(2 - x_n^2) < \frac{1}{2}(2 - x_n^2).$$

Since $2 - x_1^2 = 1$, we obtain $0 < 2 - x_n^2 \leq 2^{-(n-1)}$.

By the Archimedean property of $\mathbb{Q}$ (see the aside below), choose $m \in \mathbb{N}$ with $\frac{1}{m} < 2 - c$. Set $n = m + 1$. Since $2^m \geq m + 1 > m$, we have $2^{-(n-1)} = 2^{-m} < \frac{1}{m} < 2 - c$. Thus

$$2 - x_n^2 < 2 - c \Rightarrow c < x_n^2.$$

The positive rational x_n belongs to S , so the witnesses $a = b = x_n$ give $c \in SS$. ■

Together, the two inclusions show $SS = C_2$. Thus S represents $\sqrt{2}$.

Note. The last comparison must be $c < x_n^2$, not merely $c < x_n$: membership in SS requires a product of two positive members of S larger than c .

3.4. Aside: the Archimedean property of $\mathbb{Q}$

For every rational $\varepsilon > 0$, there is an $m \in \mathbb{N}$ such that $\frac{1}{m} < \varepsilon$. Indeed, write $\varepsilon = \frac{a}{b}$ with $a, b \in \mathbb{N}$, and take $m > b$. Then $\frac{1}{m} < \frac{1}{b} \leq \frac{a}{b}$.

This also makes $2^{-(n-1)}$ arbitrarily small: choose m with $\frac{1}{m} < \varepsilon$, and use $2^m \geq m + 1$ to get $2^{-m} < \frac{1}{m} < \varepsilon$.

4. Countability

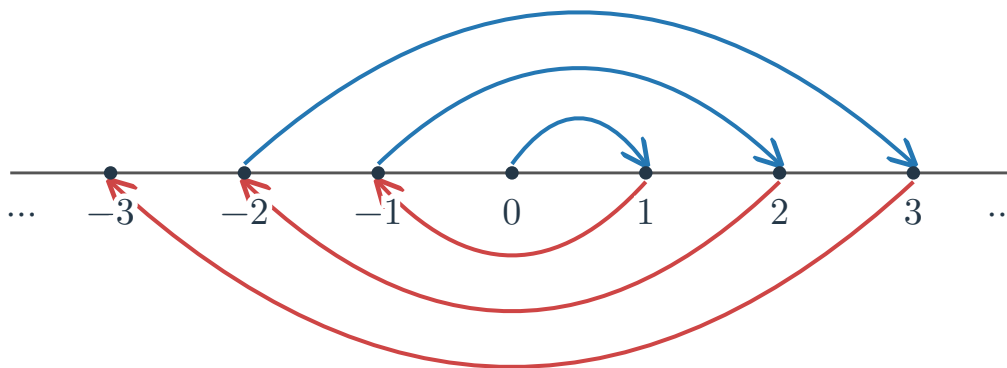
Definition 4.1. A set A is **countably infinite** if there exists a bijection between A and $\mathbb{N}$. We call a set **countable** if it is finite or countably infinite.

4.1. Enumerating $\mathbb{Z}$, $\mathbb{N} \times \mathbb{N}$, and $\mathbb{Q}$

Integers. Start at 0 and alternate between positive and negative integers:

$$0, 1, -1, 2, -2, 3, -3, \dots$$

Every integer appears exactly once. Mapping $n \in \mathbb{N}$ to the n th entry therefore gives a bijection from $\mathbb{N}$ to $\mathbb{Z}$.



Start at 0; follow the blue and red arrows alternately.

Pairs of natural numbers. List pairs of positive natural numbers by successive diagonals:

$$(1, 1), (2, 1), (1, 2), (1, 3), (2, 2), (3, 1), \dots$$

Every pair occurs on a finite diagonal, and each pair is listed once.


$$0, q_1, -q_1, q_2, -q_2, q_3, -q_3, \dots$$

Remark. The quotient of $\mathbb{N} \times \mathbb{N}$ by equality of fractions describes the positive rationals. Concatenating this enumeration with the integer bouncing pattern adds zero and both signs, giving all of $\mathbb{Q}$.

5. Power sets and uncountability

Define

$$2^A := \{\text{all subsets of } A\}.$$

5.1. Cantor's theorem and the diagonal argument

Theorem 5.1 (Cantor). *For every set A , no map $f : A \rightarrow 2^A$ is surjective. In particular, A and 2^A cannot have the same cardinality.*

Proof. Suppose $f : A \rightarrow 2^A$ is onto, and define

$$D := \{a \in A : a \notin f(a)\}.$$

Since $D \subseteq A$, we have $D \in 2^A$. Surjectivity gives a $d \in A$ such that $f(d) = D$. But

$$d \in D \Leftrightarrow d \notin f(d) \Leftrightarrow d \notin D,$$

a contradiction. ■

Note. This technique is called Russell's paradox. It is a useful trick!

The injection $a \mapsto \{a\}$ embeds A into 2^A implies that $|A| < |2^A|$. We can obtain an interesting fact about the cardinalities of the sets.

$$|\text{finite set}| < |\mathbb{N}| < |2^{\mathbb{N}}| < |2^{2^{\mathbb{N}}}| < \dots$$

Exercise 5.2. Assuming the axiom of foundation, show that no set contains itself: there is no set A with $A \in A$. **This is for someone who is interested in this type of problem. It's probably out of the scope for this course.**

5.2. Finite and infinite products

For every fixed positive integer n , the Cartesian product

$$\underbrace{\mathbb{N} \times \mathbb{N} \times \dots \times \mathbb{N}}_{n \text{ times}}$$

is countably infinite, by repeatedly applying the diagonal enumeration of $\mathbb{N} \times \mathbb{N}$.

In contrast, $\mathbb{N}^{\mathbb{N}}$, the set of all infinite sequences of natural numbers, is uncountable. It contains all sequences with entries in $\{1, 2\}$, which correspond bijectively to subsets of $\mathbb{N}$.

5.3. Comparing $\mathbb{R}$ with $2^{\mathbb{N}}$

We will show

$$|\mathbb{R}| = |2^{\mathbb{N}}|.$$

Identify each subset of $\mathbb{N}$ with its binary indicator sequence $(x_1, x_2, \dots)$, where each $x_n \in \{0, 1\}$.

Theorem 5.3 (Schröder–Bernstein). *If there is an injection $A \rightarrow B$ and an injection $B \rightarrow A$, then there is a bijection between A and B .*

Theorem 5.4. $|\mathbb{R}| = |2^{\mathbb{N}}|$

Proof. ($2^{\mathbb{N}} \subset \mathbb{R}$)

For a binary sequence $x = (x_1, x_2, \dots)$, define

$$F(x) := \sum_{n=1}^{\infty} \frac{2x_n}{3^n}.$$

This is the ternary expansion with digits 0 and 2. For example,

$$(1, 0, 0, 1, 1, \dots) \mapsto (0.20022\dots)_3.$$

The series converges because its nonnegative terms are bounded by the geometric series $\sum_{n=1}^{\infty} \frac{2}{3^n} = 1$.

Suppose $x \neq y$, and let k be the first index where they differ. Interchanging x and y if necessary, assume $x_k = 1$ and $y_k = 0$. Then

$$\begin{aligned} F(x) - F(y) &= \frac{2}{3^k} + \sum_{n=k+1}^{\infty} \frac{2(x_n - y_n)}{3^n} \\ &\geq \frac{2}{3^k} - \sum_{n=k+1}^{\infty} \frac{2}{3^n} \\ &= \frac{2}{3^k} - \frac{1}{3^k} = \frac{1}{3^k} > 0. \end{aligned}$$

Thus $F(x) \neq F(y)$, proving injectivity.

($\mathbb{R} \subset 2^{\mathbb{N}}$)

Since $\mathbb{Q}$ is countably infinite, choose a bijection $e : \mathbb{Q} \rightarrow \mathbb{N}$. Each real number is a Dedekind cut $D \subseteq \mathbb{Q}$. Define

$$G(D) := \{e(q) : q \in D\} \subseteq \mathbb{N}.$$

If $D_1 \neq D_2$, WLOG take $a \in D_1 \setminus D_2$. Then $e(a) \in G(D_1)$. Injectivity of e ensures $e(a) \notin G(D_2)$, so $G(D_1) \neq G(D_2)$.

By Schröder–Bernstein, $|\mathbb{R}| = |2^{\mathbb{N}}|$. Cantor's theorem therefore shows that $\mathbb{R}$ is uncountable. ■

Remark. There is another way to prove $\mathbb{R}$ is not countable, using the diagonal argument. Here is the classic proof I mentioned during the class.

Theorem 5.5 ($\mathbb{R}$ is uncountable). *J. Lebl [1, §1.5, Theorem 1.5.2]*

Example (Problem 5). Let $S := \{n^2 : n \in \mathbb{N}\}$. The identity function $S \rightarrow \mathbb{N}$ is not onto, since $3 \notin S$. Nevertheless, S and $\mathbb{N}$ have the same cardinality.

Proposition 5.6. *The map $f : \mathbb{N} \rightarrow S$ given by $f(n) = n^2$ is a bijection.*

Proof. **Injective:** if $a^2 = b^2$ for $a, b \in \mathbb{N}$, then $(a - b)(a + b) = 0$. Since $a + b > 0$, it follows that $a = b$.

Surjective: every $b \in S$ has the form $b = n^2 = f(n)$ for some $n \in \mathbb{N}$, by the definition of S . ■

S	1	4	9	16	...
					
$\mathbb{N}$	1	2	3	4	...

Bibliography

- [1] J. Lebl, *Basic Analysis: Introduction to Real Analysis*, vol. 1. 2026. [Online]. Available: <https://www.jirka.org/ra/realanal.pdf>